

CS 112

Introduction to Data Structures

WEEK 14

Parallelism: experiments and course recap

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This Week

- 1 Timing the matrix operations
- 2 omp parallel for
- 3 schedule(dynamic)
- 4 What we built this semester

Deck not yet written

The parallelism half of this deck has NOT been written yet. The closing recap slide below is the original and is kept.

Fourteen Weeks

Structure	Good at	Bad at
Array / vector	indexing, appending	inserting at the front
Linked list	inserting and removing anywhere you already are	indexing
Stack / queue	one disciplined end (or two)	looking in the middle
BST	ordered search, when balanced	sorted input, unless balanced
AVL tree	guaranteed $O(\lg n)$	more code, more memory
Hash table	raw lookup speed	any question about order

You built every one of these. The skill you're leaving with isn't any single structure; it's asking *which operations will this code do a million times?* before choosing.

Week 14 Recap

- A hash function turns a key into an index, $O(1)$ on average
- Collisions are inevitable; **chaining** and **open addressing** handle them
- Load factor trades memory for speed; rehashing is the doubling trick again
- Selection sort is $O(n^2)$ always; insertion sort is $O(n)$ on nearly-sorted data
- Choose the structure by the operations you repeat most

Next: Final Window · book by Dec 9 →

This deck stands on earlier CS112 materials by
Joel Adams and **Victor Norman** · adapted and extended by **Eric Araújo**

